

Test for gradient fields

$$\vec{F} = \langle P, Q, R \rangle = \langle f_x, f_y, f_z \rangle$$

If so, then

$$P_y = f_{xy} = f_{yx} = Q_x$$

$$P_z = f_{xz} = f_{zx} = R_x$$

$$Q_z = f_{yz} = f_{zy} = R_y$$

Criterion

$$\vec{F} = \langle P, Q, R \rangle$$

defined in a simply connected region is a gradient field iff

$$P_y = Q_x, \quad P_z = R_x, \quad Q_z = R_y$$

Example

$$Pdx + Qdy + Rdz$$

is exact ($= df$).

Example for which a and b is

$$axy \, dx + (x^2 + z^3) \, dy + (byz^2 - 4z^3) \, dz$$

$$P_y = ax = 2x = P_x$$

$$P_z = 0 = 0 = R_x$$

$$Q_z = 3z^2 = bz^2 = R_y$$

$$a = 2 \quad \text{and} \quad b = 3$$

Find Potential

$$f(x, y, z) = \int_C \vec{F} \cdot d\vec{r} + \text{const}$$

where C is from $(0, 0, 0)$ to (x, y, z) .

2) Antiderivatives

We want

$$f_x = 2xy,$$

$$f_y = x^2 + z^3,$$

$$f_z = 3yz^2 - 4z^3$$

Since

$$f_x = 2xy$$

then

$$f = \int 2xy \, dx = x^2y + g(y, z)$$

Now use

$$f_y = x^2 + z^3$$

so

$$x^2 + g_y = x^2 + z^3$$

Hence

$$g_y = z^3$$

Integrating with respect to y ,

$$g = yz^3 + h(z)$$

$$f_z = 3yz^2 - 4z^3$$

But

$$f = x^2y + g = x^2y + yz^3 + h(z)$$

Differentiating with respect to z ,

$$f_z = 3yz^2 + h'(z)$$

So,

$$3yz^2 - 4z^3 = 3yz^2 + h'(z)$$

Hence,

$$h'(z) = -4z^3$$

Integrating,

$$h(z) = -z^4 + C$$

Therefore,

$$f = x^2y + yz^3 - z^4 + C$$

Curl in 3D

If

$$\vec{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$$

then

$$\text{curl } \vec{F} = (R_y - Q_z)\mathbf{i} + (P_z - R_x)\mathbf{j} + (Q_x - P_y)\mathbf{k}$$

If $\vec{F}$ is defined in a simply connected region,

$$\vec{F} \text{ is conservative} \iff \text{curl } \vec{F} = 0$$

$$\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$$

We have seen

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

$$\nabla \cdot \langle P, Q, R \rangle = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

$$\text{div } \vec{F}$$

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} \\ &= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} \\ &\quad - \left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z} \right) \mathbf{j} \\ &\quad + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k} = \text{curl } \vec{F} \end{aligned}$$

Geometrically, curl measures the rotational component in a velocity field.

$$\vec{V} = \langle -\omega y, \omega x, 0 \rangle$$

$$\text{curl } \vec{V} = 2\omega \mathbf{k}$$

Lecture 31: Stokes' Theorem

Recall

$$\begin{aligned}\operatorname{curl} \vec{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} \\ &= (R_y - Q_z)\mathbf{i} + (P_z - R_x)\mathbf{j} + (Q_x - P_y)\mathbf{k} \end{aligned}$$

Measures rotational part of a velocity field.

Direction = axis of rotation

Magnitude = $2 \times$ angular velocity

$$\vec{F} = \langle a, b, c \rangle$$

$$\operatorname{curl} \vec{F} = 0$$

$$\vec{F} = \langle x, 0, 0 \rangle$$

$$\operatorname{curl} \vec{F} = 0 \quad (\text{but } \operatorname{div} \vec{F} = 1)$$

$$\vec{F} = \langle -y, x, 0 \rangle$$

$$\operatorname{curl} \vec{F} = 2\hat{k}$$

Stokes' Theorem

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dS$$

where

C = closed curve

S = any surface bounded by C

Orientation:

The orientation of C and S must be compatible.

If I walk along C in the positive direction with S to my left, then

$$\hat{n}$$

is pointing upward.

Right-hand rule:

Thumb along C in the positive direction,

index finger tangent to S ,
middle finger points in the direction of $\hat{n}$.

Comparing Stokes' Theorem to Green's Theorem

Suppose

S = portion of the xy -plane bounded by a curve C

Then

$$\oint_C \vec{F} \cdot d\vec{r} = \oint_C P dx + Q dy$$

where

$$\vec{F} = \langle P, Q, R \rangle, \quad z = 0, \quad dz = 0$$

By Stokes' Theorem,

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$$

Since

$$\hat{n} = \hat{k},$$

we obtain

$$= \iint_S (Q_x - P_y) dx dy$$

Thus Green's Theorem is a special case of Stokes' Theorem in the xy -plane.

Why Stokes' Theorem is true

- Know it for surfaces in the xy -plane.
- Also for curves C and surfaces S in any plane.
- Using that work, flux, and curl make sense independently of the coordinate system.
- Given any surface S , decompose it into tiny flat pieces.
- The sum of work around each piece equals the work along C .
- The sum of flux through each piece equals the flux through S .

Example

Work of

$$\vec{F} = z\hat{i} + x\hat{j} + y\hat{k}$$

around the unit circle in the xy -plane, counterclockwise direction:

$$\oint_C z \, dx + x \, dy + y \, dz$$

Parameterize the curve:

$$x = \cos t, \quad y = \sin t, \quad z = 0$$

Then

$$dx = -\sin t \, dt, \quad dy = \cos t \, dt, \quad dz = 0$$

Substitute:

$$\begin{aligned} \int_0^{2\pi} [0 + (\cos t)(\cos t \, dt) + 0] \\ = \int_0^{2\pi} \cos^2 t \, dt \\ = \pi \end{aligned}$$

$$z = 1 - x^2 - y^2$$

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dS$$

Compute the curl:

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & x & y \end{vmatrix} \\ &= \mathbf{i} - \mathbf{j} + \mathbf{k} = \langle 1, -1, 1 \rangle \end{aligned}$$

For the surface

$$z = f(x, y) = 1 - x^2 - y^2,$$

we have

$$\hat{n} \, dS = \langle -f_x, -f_y, 1 \rangle \, dx \, dy$$

Since

$$f_x = -2x, \quad f_y = -2y,$$

then

$$\hat{n} \, dS = \langle 2x, 2y, 1 \rangle \, dx \, dy$$

Thus,

$$\begin{aligned} \iint_S \langle 1, -1, 1 \rangle \cdot \langle 2x, 2y, 1 \rangle \, dx \, dy \\ = \iint_S (2x - 2y + 1) \, dx \, dy \end{aligned}$$

Switch to polar coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta$$

Hence,

$$\iint_S (2x - 2y + 1) \, dx \, dy = \pi$$

Lecture 32: Stokes' Theorem

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\operatorname{curl} \vec{F}) \cdot \hat{n} \, dS$$

Definition

A region is simply connected if every closed loop inside it bounds a surface.

Space with a region removed may or may not be simply connected.

Example:

$$\mathbb{R}^3 \setminus \{z\text{-axis}\}$$

is not simply connected.

Recall:

If

$$\vec{F} = \nabla f$$

is a gradient field, then

$$\operatorname{curl} \vec{F} = 0$$

Theorem

If $\vec{F}$ is defined in a simply connected region and

$$\operatorname{curl} \vec{F} = 0,$$

then $\vec{F}$ is a gradient field.

$$\int_C \vec{F} \cdot d\vec{r}$$

Proof

Assume

$$\text{curl } \vec{F} = 0$$

Then for two curves C_1 and C_2 ,

$$\int_{C_1} \vec{F} \cdot d\vec{r} - \int_{C_2} \vec{F} \cdot d\vec{r} = \oint_C \vec{F} \cdot d\vec{r}$$

where

$$C = C_1 - C_2$$

By Stokes' Theorem,

$$\iint_S (\text{curl } \vec{F}) \cdot \hat{n} dS = 0$$

Remark: Topology

Topology classifies surfaces and spaces.

The surface of a sphere is connected.

Orientability

Stokes' Theorem and surface independence

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_{S_1} (\nabla \times \vec{F}) \cdot \hat{n} dS = \iint_{S_2} (\nabla \times \vec{F}) \cdot \hat{n} dS$$

Why?

$$\begin{aligned} \iint_{S_1} (\nabla \times \vec{F}) \cdot \hat{n} dS - \iint_{S_2} (\nabla \times \vec{F}) \cdot \hat{n} dS \\ = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS \end{aligned}$$

where

$$S = S_1 - S_2$$

is a closed surface.

By the Divergence Theorem,

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS = \iiint_V \nabla \cdot (\nabla \times \vec{F}) dV$$

Check:

$$\nabla \cdot (\nabla \times \vec{F}) = 0 \quad \text{always}$$

$$R_{yx} - Q_{zx} + P_{zy} - R_{xy} + Q_{xz} - P_{yz} = 0$$

$$\nabla \cdot (\nabla \times \vec{F}) = 0$$

$$\iiint_R f \, dV$$

Rectangular coordinates:

$$dV = dz \, dx \, dy$$

Cylindrical coordinates:

$$dV = dz \, r \, dr \, d\theta$$

Spherical coordinates:

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$\iint_S \vec{F} \cdot \hat{n} \, dS$$

Formulas for $\hat{n} \, dS$

Example:

$$\hat{n} \, dS = \langle -f_x, -f_y, 1 \rangle \, dx \, dy$$

Stokes' Theorem:

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dS = \oint_C P \, dx + Q \, dy + R \, dz$$

General case (parametric surface)

If

$$z = f(x, y),$$

then

$$\hat{n} \, dS = \langle -f_x, -f_y, 1 \rangle \, dx \, dy$$

Example:

$$z = 1 - x^2 - y^2$$

gives

$$\hat{n} \, dS = \langle 2x, 2y, 1 \rangle \, dx \, dy$$

If $\vec{N}$ is a normal vector, then

$$\hat{n} \, dS = \pm \frac{\vec{N}}{|\vec{N}|} \, dx \, dy$$

$$\iint_S \vec{F} \cdot \hat{n} \, dS = \iiint_D (\operatorname{div} \vec{F}) \, dV$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dS$$

$$f(P_1) - f(P_0) = \int_C (\nabla f) \cdot d\vec{r}$$

Also, given a vector field $\vec{F}$ with

$$\operatorname{curl} \vec{F} = 0$$

then $\vec{F}$ is conservative (in a simply connected region).